

Absolute vs relative paths

Paths are how you tell the shell where to go

Absolute vs relative

- An absolute path starts from the filesystem root
- A relative path is resolved against your current working directory

Browser lab

Digital Design and Verification Platform

HomeTools

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Run pwdResolve .Resolve ..Quiz: when relative breaksResolve a file

Current directory

```
pwd
/home/lab/projects/uart_tx
```

cd ~cd ~/projectscd uart_txcd srccd ..cd ../tb

```
Starter: cwd is uart_tx - resolve ../spi_master/src

lab$ pwd · cd · ls · help
```

Resolve a path

Path to resolve

../spi_master/src

Resolve against cwd

```
relative ../spi_master/src → /home/lab/projects/spi_master/src (dir)
```

- **Absolute** — starts with `/` (from filesystem root).
- **Home** — starts with `~` (expands to `/home/lab`).
- **Relative** — everything else; joined to `pwd`, then `./` cleaned.

Lab tree

• = cwd

```
/
└─ home/
   └─ lab/
      ├── docs/
      │   └─ notes.md
      └─ projects/
          └─ spi_master/
```

Absolute vs relative paths lab

learn_unix — path abs rel

Real shell practice

```
wsl - module03-path-abs-rel/examples/paths

# Track A - real Unix
# real shell session (Track A)

$ pwd
/mnt/d/proj/designs/learning/courses/learn_unix/module03-path-abs-rel/examples/paths

$ ls -la
total 4
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 20 00:49 ..
-rwxrwxrwx 1 yongfu yongfu 1939 Jul 18 16:34 README.md

$ cd ..

$ pwd
/mnt/d/proj/designs/learning/courses/learn_unix/module03-path-abs-rel/examples

$ cd paths

$ pwd
/mnt/d/proj/designs/learning/courses/learn_unix/module03-path-abs-rel/examples/paths

$ cd ~

$ pwd
/home/yongfu
```

Real shell — absolute and relative paths

Real shell practice — try these

```
# pwd — print working directory (absolute path of “here”)
pwd
```

```
# ls -la — list this folder so you see what relative names refer to
ls -la
```

```
# cd .. — relative: move up one directory
cd ..
```

```
# pwd — confirm where you landed after the relative move
pwd
```

```
# cd paths — relative: enter this folder by name from here
cd paths
```

```
# cd ~ — go to your home directory (tilde expands to home)
```

Pitfalls to watch

- A relative path is not “wrong”, it is just tied to where you are now
- After change-directory, the same short path may fail
- Leading slash means from the root; tilde means home
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell
- When you are ready, take the short quiz, then continue to globs and wildcards